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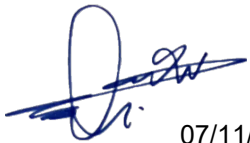
PROCEDURE FOR HANDLING OF PORTABLE HAND & POWER TOOLS

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1.0 PURPOSE

This Procedure establishes the minimum requirements for QatarEnergy-IS and Contractor portable hand tools including power activated tools, to ensure that they are operated and maintained in a safe condition.

2.0 SCOPE AND APPLICABILITY

This Procedure applies to Employees, Contractors, and Subcontractors, whether engaged directly by QatarEnergy or by others, who should perform work or services of any kind at any QatarEnergy-IS facilities and/or locations.

3.0 ROLES AND RESPONSIBILITIES

3.1 DEPARTMENT MANAGERS/TEAM LEADERS

- Ensure that this Procedure is communicated within their Department or team.
- Ensure personnel are trained and competent in the use of Portable Hand Tools.
- Ensure only authorized serviceable Portable Hand Tools are operated within their facility.

3.2 SENIOR SAFETY OFFICER / SAFETY OFFICER

- Ensure that training in the use of portable hand tools is available.
- Ensure weekly audits are conducted on portable hand tools using Attachment A.

3.3 OFFSHORE AND ONSHORE SUPERVISORS

- Ensure that this Procedure is communicated within their department or team.
- Ensure personnel are trained and competent in the use of Portable Hand Tools.
- Ensure only authorized serviceable Portable Hand Tools are operated within their facility.
- Monitor and ensure valid work permit and PPE compliance in the execution of all work where portable hand and power tools are used.

3.4 EMPLOYEE AND CONTRACTORS

- Comply with this Procedure.
- Use the correct tool for the task.
- Responsible for the inspection of Portable hand Tools to ensure they are serviceable before their use.
- Responsible for reporting any defects with Portable Hand Tools.

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4.0 PROCEDURAL STEPS

4.1 GENERAL

- All portable hand tools shall be properly maintained according to the manufacturer's instructions and subject to a formal periodic inspection (no less than annually) to ensure that tools remain in safe operating condition.
- Risk assessment shall be conducted and the appropriate permit to work (ISSoW) in place before commencing any work activities involving portable hand tools including power tools.
- Mandatory PPE shall be worn at site. Other Personal Protective Equipment (PPE) shall be worn as specified by the risk assessment, Material safety data sheet (MSDS) and Procedure 'Personal Protective Equipment Selection and Use (IS-HSE-PRC-005)'.
- No hand tool shall be used for any other purpose other than for which it was designed and purchased (Right tool for the right job).
- All portable hand tools shall be examined for any damage before and after use, by the person using the tool.
- Any tool found in an unsafe condition shall not be used, tagged "out of service" (OOS), and returned to the appropriate store for repair, replacement or disposal.
- Tools shall not be modified without both QatarEnergy and manufacturer's permission.
- Homemade tools, or job site made tools are prohibited at all QatarEnergy-IS facilities.
- Loose fitting clothing or jewellery shall not be worn, when operating any portable hand tools.
- Operators shall observe their body positioning, whilst working with portable tools to prevent injuries.
- During flange bolting, finger saving devices "Finger savers" shall be used if there is an approval to conduct flogging operations as per Procedure 'Flange Bolting (IS-FEC-PRC-234)'.
- When overhead work or working at height is required, as part of the ISSoW a risk assessment shall be used to identify the need for tool lanyards and/or other appropriate tool transport methods to be used to mitigate the possibility of dropped objects.

4.2 TRAINING

Training shall be provided in the hazards and safe-operation of portable hand tools. Training should consist of toolbox talks, risk assessments, on the job training, job safety analysis, etc and in all cases shall be documented.

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4.3 NON POWERED HAND TOOLS

4.3.1 HAMMERS AND SLEDGES

- Users shall keep their hands out of the impact area and use a tool holder "Finger Saver".
- Handles shall be non-metallic with a smooth finish and checked for splits or cracks.
- Heads of the hammer shall be firmly secured on the handle, any hammers found with loose heads shall identified as OOS and handled appropriately.
- Brass hammer only shall be used when working in Hazardous areas.

4.3.2 CHISELS AND PUNCHES

- Users shall keep chisels cutting edge sharp and in good condition.
- Chisels with "Mushroom" heads shall be identified as "out of service" (OOS) and handled appropriately.
- Users shall be instructed to strike blows away from the body.
- A vice clamp shall be used to secure small work pieces.
- Sharp or pointed tools, shall not be carried in clothing pockets.
- When chisel is not in use, the cutting edge shall be guarded to prevent damage to the edge and injuries.

4.3.3 WRENCHES

- Users shall not use a "Cheater or Snipe" to increase leverage.
- Users shall be instructed to pull on the wrench rather than push and adjust their stance to avoid a fall if the wrench slips.
- Users shall ensure the wrench fits the bolt/nut correctly.

4.3.4 SCREWDRIVERS

- Users shall use a screwdriver with the right type of blade and one that correctly fits the screw head size.
- User shall never use a bent or damage screwdriver.
- Users shall keep handles clean and in good condition.
- Users shall not place screwdrivers in clothing pockets.

4.3.5 PLIERS

- Pliers shall only be used when there are no other tools for the job.
- User shall not use pliers for cutting hardened wire unless they are specifically made to do so.
- User shall use dielectric pliers when working with electricity.
- Pliers shall be kept free from dust and grit, all movable parts lightly lubricated.
- Both hand protection and eye protection shall be worn.

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4.3.6 HAND SAWS

- Re-sharpen or replace blades that have lost cutting length.
- User shall lubricate hacksaw blades with a light machine oil to prevent heat build-up which can cause a blade to break.
- Users shall ensure saws are correctly stored away when not in use to prevent falling onto or bumping in the blade.

4.3.7 KNIVES

- Approved safety knives with self-retracting bladed shall be used.
- Kevlar gloves shall be worn at all times whilst using knives.

4.4 PORTABLE POWER TOOLS

- Portable power tools include Electrical tools, Pneumatic, Compressed Air Systems, Drills and Grinders.

4.4.1 ELECTRICAL POWER TOOLS

- All portable electrical equipment, including electrical tools shall be in compliance with and tested as per "Procedure for Control & Inspection of Electrical Equipment (IS-FOP-PRC-226).
- Electrically driven power tools shall not be operated within a potential gas environment until the area has been gas tested for explosive/toxic gases.
- Power tools shall only be used after ISSoW / PTW has been approved & issued.
- Portable electrical tools shall have a three wire cord that is free from damage and double insulated. Tools with power cord and casing damage shall be quarantined and identified as "out of service" (OOS) and handled appropriately.
- All portable electrical tools shall be plugged into an earthed receptacle or be powered by a low voltage Battery Pack.
- All portable power tools shall be fitted with a constant pressure control switch (Dead Man switch) that shall shut off the power when pressure is released.
- All portable power tools shall be inspected prior to use by the operator, to ensure no damage or defect which could lead to injury. Remove all damaged portable electric tools from use and tag them: "Do Not Use."
- All portable power tools shall be disconnected from the power, when not in use, before servicing, cleaning and when required to change accessories such as blade, bits and cutters.
- Portable power tools shall not be carried by the cord.
- Power cable shall be kept away from heat, oil, live equipment or sharp edges.
- Power tools shall not be stored in a damp place.
- Power cables shall not present a tripping hazard within the workplace.

4.4.2 PNEUMATIC POWER TOOLS

- Signs and barriers shall be erected, where pneumatic tools are being operated.
- Ensure that the compressed air supplied to the tool is clean and dry.
- Pneumatic power tools 1m tail shall be directly connected to the air tool.

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- Compressed air tool shall have a minimum operating rate of 100-110 psi and shall not exceed the manufactures recommendations.
- A safety clip and whip check shall be installed to prevent attachments from being ejected during operation.
- Tools shall be kept clean and lubricated and maintained according to the manufacturers' instructions.
- Only attachments that are recommended by the manufacture shall be used.
- Operator shall observe body position, prevent hands, feet, or body from injury.
- Operator shall reduce physical fatigue by supporting heavy tools with a counterbalance wherever possible.
- Compressed air hose shall not be used for cleaning the work area or clothing.

4.4.3 COMPRESSED AIR HOSES

- The correct fittings (Push out/In, Diameter) shall be used and jubilee clips shall not be used for hose fittings.
- Hoses designed to resist abrasion, cutting, crushing and failure shall be used.
- Hoses shall be inspected prior to use for cuts, bulges and abrasions. Tag and replace, if defective.
- Hoses shall be checked prior to use, by blowing out the air line before connecting a tool.
- All connections between air hoses shall be secured by the use of a Whip check and R clip.
- Air pressure to the hose shall be turned off and system bled down when not in use or when changing power tools.
- Pneumatic tools shall not be carried by its hose.
- Avoid creating trip hazards caused by hoses laid across walkways or curled underfoot.
- Compressed air hoses shall not be directed at personnel or used to clean coveralls

4.4.4 CHOP SAWS

- All guards shall be in place and shall be operational.
- Hands and fingers shall be kept clear of the path in which the blade travels.
- Operator shall clean the lower guard frequently to help visibility and movement. Unplug before adjusting or cleaning.
- Operators shall regularly check and tighten the blade and the blade-attachment mechanism.
- Prior to installing or changing a blade, operators shall ensure the saw is unplugged from the power source.
- Operators shall ensure that the blade and its related washers and fasteners are correctly positioned and secured on the saw's arbor.
- To avoid losing control or placing hands in the blade path, hold or clamp all material securely against the fence when cutting. Do not perform operations freehand.
- Use the brake if one is provided. To avoid contact with a coasting blade, do not reach into the cutting area until the blade comes to a full stop.

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- After completing a cut, release the trigger switch and allow the blade to come to a complete stop, and then raise the blade from the work piece.

4.4.5 GRINDERS INCLUDING ABRASIVE DISCS

Common causes of injury to operators and nearby workers from angle grinders are lacerations from attachments that break and become projectiles, and lacerations from angle grinder kickback.

- Operators shall have training in the operation of portable disc grinder including abrasive and cutting discs.
- Before using the disc grinder, examine the power cord for any damage.
- Before starting the grinder ensure that the grinding wheel, guard and attachment are secure and correctly fitted.
- Ensure the correctly graded disc is used for the material which is to be worked
- Check the wheel carefully for cracks or damage before starting work.
- All portable grinders shall be fitted with a constant pressure control switch (Dead Man switch) that shall shut off the power when pressure is released.
- Grinders shall be operated with both hands, this will improve control and prevent potential injury.
- Damaged grinders or grinding discs shall not be used and shall be removed for repair or disposal.
- Secure and support the work piece using clamps or vices. Ensure fingers, hands and power cable are clear of the grinding wheel.
- Ensure the grinder reaches the operating speed before using it on the work piece.
- Do not put excess pressure on the wheel, as it can cause the wheel to shatter.
- Be aware of flying sparks. Hold the grinder so that sparks fly away from you, other persons and combustible material.
- Never use cutting discs for grinding purposes. Cutting discs shall not be subjected to any lateral pressure.
- Ensure that the accessories used always fulfil the maximum grinding-disc diameter minimum requirements, diameter of the central hole in the grinding disc shall fit the inner flange without any play.
- Use clamps or other practical way to secure and support the work piece to a stable platform.
- Do not hold work by hand or against your body.
- Abrasive wheels shall be stored and handled with care in accordance with manufacturer's instructions.
- Disconnect the plug from the power source before making any adjustments, changing accessories, or storing the tool.
- Inspect the tool before each use to ensure cutting edges are sharp, the on and off switch works, the clamp nut is in place and secure,
- Do not leave the grinder running when not in use.
- Never use your feet to hold down materials for cutting or grinding.

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4.5 MACHINE GUARDS

One or more methods of machine guarding shall be provided to protect the Operator and other employees in the machine area from hazards such as those created by point of operation, nip points, rotating parts, flying chips and sparks.

Examples of guarding methods are barrier guards, two hand tripping devices and electric safety devices.

QatarEnergy-IS shall ensure that risk is As Low As Reasonably Practicable (ALARP) and guards are affixed to machinery and the moving parts

The guard shall be such that it does not offer an accident hazard in itself.

These include but are not restricted to:

- Fixed plant and equipment where the guarding safeguard is part of the design.
- Hand tools, pedestal grinders and drills with rotating parts.
- Rented mobile plant machinery and equipment.

No machinery guarding designed for the protection of contact or impact with moving parts is to be removed except under ISSOW/ PTW conditions for work to be performed on the equipment.

Any work / task that require the guarding to be removed with the machinery operating (e.g. dynamic balancing of motors) shall be risk assessed and the task participant made aware of the potential entanglement hazards and the mitigation for those hazards.

5.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/ External Document	Document Referencing
Standard For Hand Protection	CORP-SAF-STD-017	Internal	Upward
IS HSE Training Procedure	IS-HSE-PRC-002	Internal	Sideways
Personal Protective Equipment Selection and Use	IS-HSE-PRC-005	Internal	Sideways
IS Integrated Safe System of Work - ISSOW	IS-FOP-PRC-100	Internal	Sideways
IS Control & Inspection of Electrical Equipment	IS-FOP-PRC-226	Internal	Sideways
IS Electrical Safety Regulations	IS-FOP-PRC-500	Internal	Sideways
IS Flange Bolting Procedure	IS-FEC-PRC-234	Internal	Sideways

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6.0 DEFINITIONS

TABLE 2: KEY TERMS DEFINED

Term	Definition
Cheater / Snip Wrench	A type of torque multiplier wrench.
Dead Man Switch	Constant pressure control switch
Kickback	When a grinder or rotating device suddenly thrusts back towards the operator as a result of it grabbing or jamming on the materials being work on.

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APPENDIX**APPENDIX A - HAND TOOL AUDIT**

Hand Tool Inspection				
Location:	Area:	Name:	Signature:	Date:
Observations		Tick Appropriate Box		Comments:
		YES	NO	
General				
Hand tools are in good condition.				
The correct hand tool is being used for the task.				
The worksite is free from homemade fabricated or modified tools.				
The correct PPE is worn based on the PTW / SDS.				
Electrical Hand Tools				
All portable electrical hand tools, are being used within the requirements of the ISSOW / WCC.				
All portable electrical tools are marked with an appliance number and listed in the portable equipment list.				
All portable electrical tools have been inspected within the last 6 months as per IS-FOP-PRC-500.				
Safety guards are fitted.				
Constant pressure control switch (Dead Man switch) is fitted and working.				
Remedial Action (If Applicable):				
Auditor Name:		Signature:	Job Title:	Date Completed:

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APPENDIX B - REVISION HISTORY LOG

TABLE 3: REVISION HISTORY LOG

Item Revised	Revision Description	Page No.
Common	Updated based on new format for management system documents	-
Remarks:		

APPENDIX C- LIST OF ABBREVIATIONS

TABLE 4: LIST OF ABBREVIATIONS

Term	Abbreviations
ALARP	As Low as Reasonably Practicable
ISSOW	Integrated Safe System of Work (PTW system)
OOS	Out of Service
PTW	Permit to Work
MSDS	Material Safety Data Sheet